Seonho Lee

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U.S. work authorization: H-1B

GPU and system software engineer working on Apple Intelligence and Private Cloud Compute. Focused on LLM inference, ML frameworks, runtime performance, and efficient AI infrastructure.

EXPERIENCE

Apple Jun 2025 – Present
Software Engineer Cambridge, MA

- Develop GPU and system software for Apple Intelligence and Private Cloud Compute (PCC) on Apple Silicon.
- Optimize LLM inference performance across the GPU and system software stack.

AMD May 2024 – Aug 2024
Research Intern Bellevue, WA

- Analyzed GPU power and performance trade-offs in LLM training across workload characteristics and GPU clock frequencies.
- Studied compute-communication overlap to identify opportunities for more energy-efficient distributed training.

Georgia Institute of Technology Aug 2023 – May 2024
Graduate Research Assistant Atlanta, GA

- Developed NeuSight to predict deep learning training and inference performance on unseen GPUs and workloads; implemented prediction, multi-GPU, evaluation, and reproducibility components.
- Characterized performance and power effects of compute-communication overlap in distributed deep learning training.

KAIST Jun 2021 – May 2022
Research Intern Daejeon, South Korea

- Co-designed HAMMER, a self-attention hardware accelerator; built a C++ cycle-level simulator, developed Verilog RTL, and performed synthesis for power and area estimation.

TECHNICAL SKILLS

Languages: C, C++, Objective-C++, Swift, Python, CUDA, SystemVerilog
Frameworks and tools: Metal, PyTorch, CUDA Toolkit, NumPy, Pandas, Git, Linux, Docker

EDUCATION

Georgia Institute of Technology May 2025
M.S., Electrical and Computer Engineering | GPA: 4.0/4.0

KAIST Feb 2023
B.S., Electrical Engineering | GPA: 4.0/4.3

SELECTED PUBLICATIONS

Forecasting GPU Performance for Deep Learning Training and Inference
Seonho Lee, Amar Phanishayee, Divya Mahajan
ASPLOS 2025 | Selected for IEEE Micro Top Picks 2026
Paper: https://dl.acm.org/doi/10.1145/3669940.3707265 | Code: https://github.com/scai-tech/NeuSight

Characterizing Compute-Communication Overlap in GPU-Accelerated Distributed Deep Learning: Performance and Power Implications
Seonho Lee, Jihwan Oh, Seokjin Go, Divya Mahajan
ISPASS 2025 (Poster)
Paper: https://arxiv.org/abs/2507.03114

HAMMER: Hardware-friendly Approximate Computing for Self-attention with Mean-redistribution and Linearization
Seonho Lee, Ranggi Hwang, Jongse Park, Minsoo Rhu
IEEE Computer Architecture Letters, 2023
Paper: https://ieeexplore.ieee.org/document/10005793